The spectral localiser via E-theory

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Based on joint work with Bram Mesland (Leiden):

- *The odd spectral localiser via asymptotic morphisms and quasi-projections.* Ann. K-Theory 11, no. 2 (2026), 213—260.

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$$\mathrm{K}_1(A) \times \mathrm{KK}_1(A, B) \to \mathrm{K}_0(B)$$

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in a “nice” way

. . . that is “physically / numerically applicable”.

The odd index pairing, 1

Setup: A a unital C*-algebra, $\mathcal{A} \subseteq A$ a dense *-subalgebra;

- $v \in \mathcal{A}$ a unitary;
- $(\mathcal{A}, \mathcal{H}, D)$ an odd (=ungraded) spectral triple over A;

Recall that this means:

- $A \subseteq \mathbb{B}(\mathcal{H})$, D is an unbounded, self-adjoint operator on $\mathcal{H}$ with compact resolvent, $[D, a]$ is bounded for all $a \in \mathcal{A}$.
- D may be thought of as a first-order differential operator or a Dirac operator.

Example $(C^\infty(M), L^2(M, S), D)$ for a compact spin^c manifold with spinor bundle S and Dirac operator D.

The odd index pairing, 2

- v gives $[v] \in \mathrm{K}_1(A)$;
- $(\mathcal{A}, \mathcal{H}, D)$ gives $[D] \in \mathrm{KK}_1(A, \mathbb{C})$.

The odd index pairing is the Kasparov product

$$\mathrm{K}_1(A) \times \mathrm{KK}_1(A, \mathbb{C}) \to \mathrm{K}_0(\mathbb{C}) \simeq \mathbb{Z}, \quad ([v], [D]) \mapsto \mathrm{ind}(\underbrace{PvP + 1 - P}_{\text{Fredholm}}),$$

where P is the positive spectral projection of D:

$$P := \chi_{[0,+\infty)}(D).$$

Odd index pairing on the circle

- $\mathbb{T} := \left\{ e^{2\pi i \vartheta} \,\middle|\, \vartheta \in [0,1) \right\}$.
- $v \in C^1(\mathbb{T})$ a unitary function.
- $(\mathcal{A}, \mathcal{H}, D) = \left(C^1(\mathbb{T}), L^2(\mathbb{T}), -\frac{i}{2\pi}\frac{d}{d\vartheta} \right)$ an odd spectral triple.
- $T_v := PvP + 1 - P$ is a Fredholm operator on $P\mathcal{H} = H^2(\mathbb{T})$, where

$$H^2(\mathbb{T}) := \left\{ f \in L^2(\mathbb{T}) \,\middle|\, f = \sum_{n \geq 0} a_n e^{2\pi i n \vartheta} \right\} \simeq \ell^2(\mathbb{N}).$$

Theorem (Gohberg–Krein) $\quad \operatorname{ind}(T_v) = -\operatorname{wind}(v)$, where

$$\operatorname{wind}(v) := \frac{1}{2\pi i} \int_{\mathbb{T}} \frac{v'(\vartheta)}{v(\vartheta)} d\vartheta.$$

This is the topological invariant for the Su–Schrieffer–Heeger model (of the polyacetylene).

The odd spectral localiser, 1

Loring and Schulz-Baldes have introduced a method called spectral localiser to compute the odd index pairing $[v] \times [D]$.

Input:

- an odd spectral triple $(\mathcal{A}, \mathcal{H}, D)$;
- a unitary $v \in \mathcal{A}$ (or more generally, an invertible);

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- a unitary $v \in \mathcal{A}$ (or more generally, an invertible);

Output:

- a (family of) unbounded self-adjoint operators $(L_\kappa)_{\kappa>0}$, where
$$L_\kappa := \begin{pmatrix} \kappa D & v \\ v^* & -\kappa D \end{pmatrix}$$
parametrised by a "tuning parameter" $\kappa > 0$;
- a (family of) finite-dimensional matrices $L_{\kappa,\lambda}$ which are truncations of L_κ onto a spectral subspace $\mathcal{H}_\lambda$;
- half-signature $\frac{1}{2}\operatorname{sig}(L_{\kappa,\lambda})$.

The odd spectral localiser, 2

Theorem (Loring–Schulz-Baldes) For:

- sufficiently small tuning parameter κ and
- sufficiently large spectral threshold λ, then

$$\boxed{\operatorname{ind}(PvP + 1 - P) = \frac{1}{2}\operatorname{sig}(L_{\kappa,\lambda})}$$

where $L_{\kappa,\lambda}$ is the truncation of L_κ onto the spectral subspace

$$\mathcal{H}_\lambda := \chi_{|x|\leq\lambda}(D \oplus D)(\mathcal{H} \oplus \mathcal{H}).$$

(finite-dimensional, if D has discrete spectrum)

- Computation with finite spectral data $\Longleftrightarrow$ finite energy resolution / spatial scale.
- Optimal for physical measurement / numerical simulation.

The odd spectral localisers, 3

Enigmas:

1. What *is* the spectral localiser, indeed?
2. What happens in a “spectral truncation”?
3. What is the half-signature of $L_{\kappa,\lambda}$?

The odd spectral localisers, 3

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1. What *is* the spectral localiser, indeed?
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3. What is the half-signature of $L_{\kappa,\lambda}$?

Short answers:

1. It *is* the Kasparov product, computed with an asymptotic morphism generated by a spectral triple / unbounded Kasparov module.
2. Homotopy invariance + additivity of K-theory.
3. The half-signature realises the isomorphism $\mathrm{K}_0(\mathbb{K}) \simeq \mathbb{Z}$

I shall answer in the order 3 $\Rightarrow$ 2 $\Rightarrow$ 1 $\Rightarrow$ 2 $\Rightarrow$ 3.

Signature versus rank

Let $\mathcal{K}$ be a finite-dimensional Hilbert space. Then $\mathbb{K}(\mathcal{K}) \simeq \mathbb{M}_{\dim \mathcal{K}}(\mathbb{C})$.

- The signature of a self-adjoint matrix L is

$$\begin{aligned}\mathrm{sig}(L) &= \#(\text{pos. eigenvalues of } L) - \#(\text{neg. eigenvalues of } L)\\ &= \mathrm{rank}(\chi_{(0,+\infty)}(L)) - \mathrm{rank}(\chi_{(-\infty,0)}(L)).\end{aligned}$$

- If p is a projection on $\mathcal{K}$, then

$$\begin{aligned}\mathrm{sig}(2p - \mathrm{id}_{\mathcal{K}}) &= \mathrm{rank}(p) - \mathrm{rank}(\mathrm{id}_{\mathcal{K}} - p)\\ &= 2\,\mathrm{rank}(p) - \dim \mathcal{K}.\end{aligned}$$

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- If p is a quasi-projection on $\mathcal{K}$, then

$$\begin{aligned}\operatorname{sig}(2p - \mathrm{id}_{\mathcal{K}}) &= \operatorname{rank}(\kappa_0(p)) - \operatorname{rank}(\kappa_0(\mathrm{id}_{\mathcal{K}} - p)) \\ &= 2\operatorname{rank}(\kappa_0(p)) - \dim \mathcal{K}.\end{aligned}$$

$$\kappa_0 : \mathbb{C} \setminus \{z \in \mathbb{C} \mid \operatorname{Re} z = 1/2\} \to \{0,1\}, \quad \kappa_0(z) := \begin{cases} 1, & \operatorname{Re} z > 1/2; \\ 0, & \operatorname{Re} z < 1/2. \end{cases}$$

K-theory via quasi-projections and quasi-idempotents, 1

Quasi-projection: $\|p^2 - p\| < 1/4$, $p = p^*$;
Quasi-idempotent: $\|e^2 - e\| < 1/4$.

- A unital C*-algebra. Then (see e.g. [Willett–Yu])

$$\begin{aligned} \mathrm{K}_0(A) &= \mathrm{Gr}(\text{homotopy classes of projections in } \mathbb{M}_\infty A) \\ &= \mathrm{Gr}(\text{homotopy classes of quasi-projections in } \mathbb{M}_\infty A). \end{aligned}$$

- Class of quasi-projection p = class of the projection $\kappa_0(p)$.

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- Class of quasi-projection p = class of the projection $\kappa_0(p)$.
- $\mathcal{K}$ finite-dimensional, $p \in \mathbb{K}(\mathcal{K} \oplus \mathcal{K})$ a quasi-projection: $\mathrm{K}_0(\mathbb{K}(\mathcal{K} \oplus \mathcal{K})) \xrightarrow{\sim} \mathbb{Z}$ maps $[p]$ to

$$\mathrm{rank}(\kappa_0(p)) = \frac{1}{2}\,\mathrm{sig}(2p - \mathrm{id}_{\mathcal{K}\oplus\mathcal{K}}) + \frac{1}{2}\dim(\mathcal{K} \oplus \mathcal{K}).$$

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hence maps

$$[p] - \left[\begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}\right] \longmapsto \tfrac{1}{2}\,\mathrm{sig}(2p - \mathrm{id}_{\mathcal{K}\oplus\mathcal{K}}).$$

K-theory via quasi-projections and quasi-idempotents, 2

What is good about quasi-projections/-idempotents?

- Quasi-projections/-idempotents form open sets as opposed to projections/idempotents, hence allow more homotopies.

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- If both PeP is close enough to a projection, then e is homotopic (as a quasi-projection!) to $PeP + (1-P)e(1-P)$.
- This would imply

$$[e] = [PeP] + [(1-P)e(1-P)]$$

as classes in K-theory.

Asymptotic morphism, 1

Definition An asymptotic morphism $A \dashrightarrow B$ is a family of maps $(\Phi_t : A \to B)_{t \in [1,\infty)}$ such that:

1. $t \mapsto \Phi_t(a)$ is continuous for all $a \in A$;
2. for all $a, a' \in A$ and $\lambda \in \mathbb{C}$,

$$\left.\begin{array}{c} \Phi_t(a^*) - \Phi_t(a)^* \\ \Phi_t(aa') - \Phi_t(a)\Phi_t(a') \\ \Phi_t(a + \lambda a') - \Phi_t(a) - \lambda\Phi_t(a') \end{array}\right\} \to 0 \quad \text{as } t \to \infty.$$

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An asymptotic morphism $A \dashrightarrow B$ represents a class in $\mathrm{E}(A, B)$, the E-theory group of Connes and Higson.

Asymptotic morphism, 2

- E-theory is another bivariant K-theory, characterised by different universal properties from Kasparov's KK-theory.
- By universal property, there is a natural transformation (in both variables A and B):
$$\mathrm{KK}_1(A,B) \to \mathrm{E}_1(A,B)$$
which is a natural isomorphism if A is nuclear.
- In particular, the following diagram commutes:

$$\begin{tikzcd}
\mathrm{K}_1(A) \arrow[r, "\mathrm{KK}_1(A,B)"] \arrow[d, "\mathrm{ind}"] \arrow[dr, phantom, "\Downarrow"] & \mathrm{K}_0(B) \arrow[d, "\pm 1"] \\
\mathrm{K}_0(SA) \arrow[r, "\mathrm{E}_1(A,B)"'] & \mathrm{K}_0(B).
\end{tikzcd}$$

- We may pass to $\mathrm{E}_1(A,B)$ to compute the odd index pairing.

Asymptotic morphism, 3

- Roughly: $\mathrm{E}(A,B) =$ "homotopy equivalence classes of asymptotic morphisms $A \dashrightarrow B$".
- We hope to find a useful representative for the image of $[D] \in \mathrm{KK}_1(A,B)$ in $\mathrm{E}_1(A,B)$.

Theorem (Higson–Kasparov 01; L–Mesland 25)

Let $(\mathcal{A}, E, D)$ be an essential odd unbounded Kasparov A-B–module. The asymptotic morphism

$$\Phi_t^D : \mathrm{C}_0(\mathbb{R}) \otimes A \to \mathbb{K}_B(E), \quad f \otimes a \mapsto f(t^{-1}D)a,$$

represents the class of the image of $[D] \in \mathrm{KK}_1(A,B)$ in $\mathrm{E}_1(A,B)$ under the natural transformation $\mathrm{KK}_1 \Rightarrow \mathrm{E}_1$.

Here being essential means $\overline{\mathcal{A}E} = E$.

Asymptotic morphism, 4

We have $\Phi_t^D([v]) = \langle [v], [D] \rangle$ is the index pairing (Kasparov product). The left-hand side is compute as follows.

1. The image of $[v] \in \mathrm{K}_1(A)$ in $\mathrm{K}_0(SA)$ is given by $[e] - [f]$, where

$$e := \begin{pmatrix} s(x)^2 \otimes 1 & c(x)s(x) \otimes v \\ c(x)s(x) \otimes v^* & c(x)^2 \otimes 1 \end{pmatrix}, \quad f := \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}.$$

   and $c, s \colon \mathbb{R} \to [0,1]$ are any choices of continuous functions satisfying

$$c^2(x) + s^2(x) = 1, \quad \lim_{x\to\infty} s(x) = 1, \quad \lim_{x\to-\infty} c(x) = 1.$$

   Proof. K-theory boundary map for $SA \rightarrowtail CA \twoheadrightarrow A$. $\square$

Asymptotic morphism, 5

2. For $t \gg 0$, $\Phi_t^D([e]-[f]) := [\Phi_t^D(e)] - [\Phi_t^D(f)]$, where

$$\Phi_t^D(e) = \underbrace{\begin{pmatrix} s_t^2 & c_t s_t v \\ c_t s_t v^* & c_t^2 \end{pmatrix}}_{\text{quasi-idempotent}}, \quad \Phi_t^D(f) = \underbrace{\begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}}_{\text{projection}}.$$

$\left(c_t := c(t^{-1}D) \text{ and } s_t := s(t^{-1}D)\right)$.

Asymptotic morphism, 5

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$\left(c_t := c(t^{-1}D)\right.$ and $\left. s_t := s(t^{-1}D)\right)$.

3. Up to small perturbation (which happens if certain commutator are bounded!), we may replace them by quasi-projections

$$e_t := \underbrace{\begin{pmatrix} s_t^2 & \sqrt{c_t s_t}\, v \sqrt{c_t s_t} \\ \sqrt{c_t s_t}\, v^* \sqrt{c_t s_t} & c_t^2 \end{pmatrix}}_{\text{quasi-projection}}, \quad f_t := \underbrace{\begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}}_{\text{projection}}.$$

Then $[e_t] - [f_t] = \langle [v], [D] \rangle$.

Spectral truncation, 1

Setup:

- B a C*-algebra;
- $\mathcal{E}$ a Hilbert B-module;
- $\mathcal{D}$ an unbounded, self-adjoint and regular operator on $\mathcal{E}$;
- λ a positive real number ("spectral threshold").

Definition A spectral decomposition of $\mathcal{E}$ is a pair of mutually complemented submodules $(\mathcal{E}_\lambda^\downarrow, \mathcal{E}_\lambda^\uparrow)$ of $\mathcal{E}$, such that

$$\langle \mathcal{D}\xi, \mathcal{D}\xi\rangle \geq \lambda^2 \langle \xi, \xi\rangle \quad \text{for all } \xi \in \operatorname{dom}\mathcal{D} \cap \mathcal{E}_\lambda^\uparrow.$$

Remark For any $\lambda > 0$, a spectral decomposition $\mathcal{E} = \mathcal{E}_\lambda^\downarrow \oplus \mathcal{E}_\lambda^\uparrow$ always exists if $B = \mathbb{C}$; may not exist for general B.

Spectral truncation, 2

Assumption $(E^{\downarrow}_{\lambda}, E^{\uparrow}_{\lambda})$ is a spectral decomposition of $\widehat{E} := E \oplus E$ for the operator $D \oplus D$.

Then e_t decomposes as

$$e_t = \begin{pmatrix} p^e_{t,\lambda} & {m^e_{t,\lambda}}^* \\ m^e_{t,\lambda} & q^e_{t,\lambda} \end{pmatrix} \curvearrowright \begin{pmatrix} E^{\downarrow}_{\lambda} \\ E^{\uparrow}_{\lambda} \end{pmatrix} \begin{matrix} \text{“lower” submodule} \\ \text{“upper” submodule} \end{matrix}$$

Lemma For $t \gg 0$ and $\lambda \gg 0$, the following quasi-projections are homotopic:

$$\begin{pmatrix} p^e_{t,\lambda} & {m^e_{t,\lambda}}^* \\ m^e_{t,\lambda} & q^e_{t,\lambda} \end{pmatrix} \quad \text{and} \quad \begin{pmatrix} p^e_{t,\lambda} & \\ & q^e_{t,\lambda} \end{pmatrix}.$$

Therefore, as a class in $K_0(\mathbb{K}_B(E)^+)$:

$$[e_t] = [p^e_{t,\lambda}] + [q^e_{t,\lambda}].$$

Spectral truncation, 3

We have $[f_t] = [p^f_{t,\lambda}] + [q^f_{t,\lambda}]$, and

$$q^e_{t,\lambda} \sim q^f_{t,\lambda}$$

are asymptotically unitarily equivalent. So $[q^e_{t,\lambda}] = [q^f_{t,\lambda}]$.

Summing up:

$$\begin{aligned} [e_t] - [f_t] &= [p^e_{t,\lambda}] + [q^e_{t,\lambda}] - [p^f_{t,\lambda}] - [q^f_{t,\lambda}] \\ &= [p^e_{t,\lambda}] - [p^f_{t,\lambda}], \end{aligned}$$

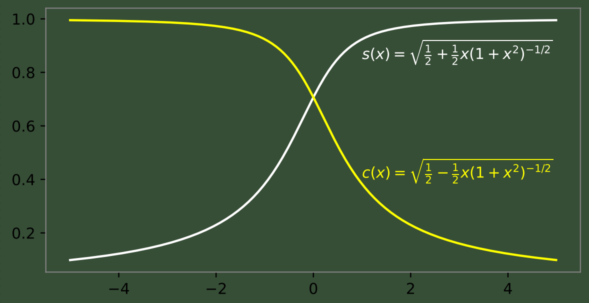
both $p^e_{t,\lambda}$ and $p^f_{t,\lambda}$ are quasi-projections on $E^{\downarrow}_{\lambda}$.

If $(\mathcal{A}, E, D)$ is a spectral triple and D has discrete spectrum, then $p^e_{t,\lambda}$ and $p^f_{t,\lambda}$ are finite-dimensional quasi-projections.

Emergence of the spectral localiser, 1

A choice of $c(x)$ and $s(x)$:

$$c(x) := \sqrt{\frac{1}{2} - \frac{1}{2}x(1+x^2)^{-1/2}}, \quad s(x) := \sqrt{\frac{1}{2} + \frac{1}{2}x(1+x^2)^{-1/2}}.$$



They satisfy the prescribed commutator estimates.

Emergence of the spectral localiser, 2

- $2p^{e}_{t,\lambda} - 1$ is given by ($D_t := t^{-1}D$):

$$\begin{pmatrix} D_t(1+D_t^2)^{-1/2} & (1+D_t^2)^{-1/4} v (1+D_t^2)^{-1/4} \\ (1+D_t^2)^{-1/4} v^* (1+D_t^2)^{-1/4} & -D_t(1+D_t^2)^{-1/2} \end{pmatrix}$$

$$= \begin{pmatrix} (1+D_t^2)^{-1/4} & 0 \\ 0 & (1+D_t^2)^{-1/4} \end{pmatrix} \underbrace{\begin{pmatrix} D_t & v \\ v^* & -D_t \end{pmatrix}}_{L_{t^{-1},\lambda}} \begin{pmatrix} (1+D_t^2)^{-1/4} & 0 \\ 0 & (1+D_t^2)^{-1/4} \end{pmatrix}.$$

- $2p^{f}_{t,\lambda} - 1$ is given by

$$\begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix}$$

and has no signature.

Emergence of the spectral localiser, 3

Theorem (Loring–Schulz-Baldes 17, 19; L–Mesland 25)

Assume that $(E_\lambda^{\downarrow}, E_\lambda^{\uparrow})$ is a spectral decomposition of $\widehat{E} := E \oplus E$ (with spectral threshold λ). Define

$$L_{t^{-1}} := \begin{pmatrix} t^{-1}D & v \\ v^* & -t^{-1}D \end{pmatrix},$$

and $L_{t^{-1},\lambda}$ be its truncation onto $E_\lambda^{\downarrow}$. Let:

- $\varepsilon, \delta > 0$ satisfy $\varepsilon + \delta < \frac{1}{400}$;
- t, λ satisfy $t > 4\varepsilon^{-1}\|[D, v]\|$ and $\lambda > t\delta^{-1}$,

Then

$$\frac{1}{2}\operatorname{sig}(L_{t^{-1},\lambda}) = \langle [v], [D] \rangle \in \mathrm{K}_0(\mathbb{K}_B(E)).$$

Further remarks

Why unbounded Kasparov modules?

The data of an unbounded Kasparov module yield several commutator estimates that are used to construct:

- homotopies of quasi-projections/idempotents;
- homotopies of asymptotic morphisms.

Numerical index pairings with traces

The spectral truncation technique also applies to computing index pairings associated to certain semi-finite spectral triples, arising as

$$\mathrm{K}_1(A) \xrightarrow{\times[D]} \mathrm{K}_0(B) \xrightarrow{\tau_*} \mathbb{R},$$

and we recover the so-called semi-finite spectral localiser of Schulz-Baldes and Stoiber.

Summary

- The index pairing $\langle [v], [D] \rangle$ may be computed with asymptotic morphisms.
- This yields quasi-projection representatives of K-theory.
  A spectral truncation may be applied, providing the existence of a suitable spectral decomposition.
- The spectral localiser occurs as such an index pairing. Similar story for its semi-finite analogue.
- Outlook: relation to spectral flow and operator systems.